



Modern MLOps

LECTURE

MLOps on Databricks



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The lecture **MLOps on Databricks** introduces DataOps practices, SQL, and workflows, highlighting tools, jobs, and DABs for efficient data and ML operations.

DataOps tasks and tools in Databricks

The table lists common DataOps tasks and tools in Databricks:

Ingest & transform data	Autoloader and Apache Spark*
Track data changes to including versioning & lineage	Delta tables*
Build, manage, & monitor data processing pipelines	Delta Live Tables*
Ensure data security & governance	Unity Catalog*
Exploratory data analysis, dashboards, & general coding	Databricks SQL, Dashboards, and Databricks notebooks
Schedule data pipelines & Automate general workflows	Databricks Workflows
Create, store, manage, & discover features	Databricks Feature Store*
Data monitoring	Lakehouse Monitoring**

* Discussed in other associated machine learning courses.

** Covered later in the training

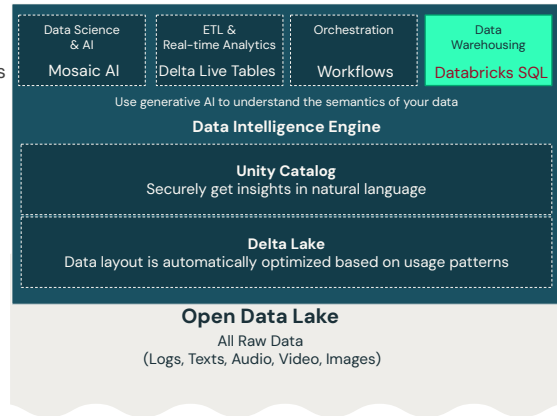
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- **Good ML models depend on reliable data pipelines and infrastructure.**
- With the **Databricks Data Intelligence Platform**, the entire data pipeline from ingesting data to the outputs from the served model is on a single platform and uses the same toolset,
- **The table lists common DataOps tasks and tools in Databricks:**

Databricks SQL

Delivering analytics on the freshest data with data warehouse performance and data lake economics

- Better price / performance than other cloud data warehouses
- Simplify discovery and sharing of new insights
- Connect to familiar BI tools, like Tableau or Power BI
- Simplified administration and governance



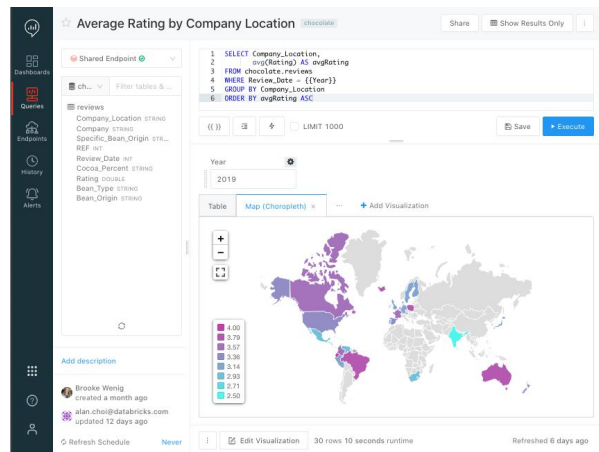
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- **Databricks SQL** is a serverless data warehouse on the Databricks Lakehouse Platform. It allows you to **run all your SQL and BI applications** at scale
 - Built on Lakehouse architecture and utilizing **Unity Catalog** for unified **security** and **governance** ensuring consistency in security and compliance.
 - **Integrate with BI tools**.... Ex: Databricks Dashboard, Tableau, Hex and Power BI etc.
- **Use Cases and Performance Enhancements**
 - Ideal for handling large-scale **ETL processes**, **real-time analytics**, and **machine learning workflows**.
 - Use **materialized views** to greatly improve query performance and reduce operational costs for data-intensive applications.

Databricks SQL is a Databricks service designed with SQL analysts in mind. It provides better price / performance, and it simplifies discovery and sharing of new insights through BI tools with which you are already familiar, like Tableau and Power BI. Finally, Databricks SQL simplifies administration and governance.

A new home for Data Analysts

- Enable data analysts to quickly **perform ad-hoc and exploratory data analysis**, with a new SQL query editor, visualizations and dashboards.
- **Automatic alerts can be triggered** for critical changes, allowing to respond to business needs faster.



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Databricks SQL enhances analytics by combining powerful query capabilities with visualization and collaboration features. Input widgets enable parameterization in notebooks and dashboards, while User-Defined Functions (UDFs) in SQL and Python allow custom transformations, and AI Functions integrate large language models directly into queries. The platform supports a wide range of built-in operators and functions for diverse data types and transformations, from strings and arrays to JSON and timestamps. With a full-featured SQL-native editor, analysts can write queries in familiar syntax, explore Delta Lake schemas, save reusable snippets, and benefit from query result caching for faster performance. Insights can be quickly transformed into rich visualizations and organized into dashboards through an intuitive drag-and-drop interface. These dashboards can be shared across teams or externally via browser access, set to auto-refresh, and configured with alerts to notify stakeholders of important data changes—all while continuing to integrate smoothly with existing BI tools.

DataOps Tasks and Tools in Databricks

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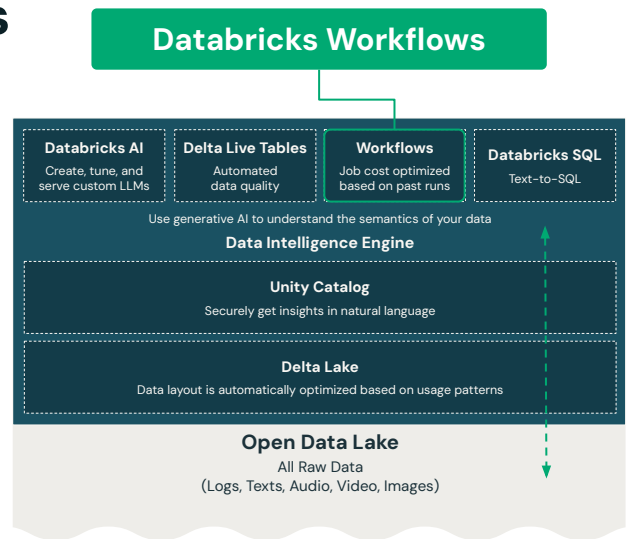
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Databricks Workflows

- Workflows is a **fully-managed cloud-based general-purpose task orchestration service** for the entire Lakehouse.
- Workflows is a service for **data engineers, data scientists and analysts** to build reliable data, analytics and AI workflows on any cloud.



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- **Databricks Workflows = A comprehensive task orchestration service** offered as part of the Databricks Lakehouse platform.
 - Suited for orchestrating comprehensive workflows that may include **data ingestion, transformation, and analytical processing**.
 - i. supporting: **data engineers, scientists, analysts, etc.** in deploying dependable workflows across any cloud infrastructure.

Let's start with a short introduction of Databricks Workflows.

Workflows is a **fully-managed, cloud-based, general-purpose task orchestration service** for the entire platform.

Workflows is a service for data engineers, data scientists and analysts to build reliable data, analytics and AI workflows on any cloud.

Databricks Workflows enables all data teams to orchestrate any combination of tasks, such as notebooks, SQL, ML models and python code. Furthermore, as Workflows sits on the top of the Databricks platform, it can be fully integrated with other services. This allows data teams to use data governance and monitoring services.

With Databricks Workflows, your team can easily create, run, monitor and repair data pipelines without managing any infrastructure.

Databricks Workflows

Databricks has two main task orchestration services:

Job Workflows			Delta Live Tables
- Execute jobs on a predefined schedule - Series of interrelated tasks.	Perform machine learning operations by running tasks within job frameworks like MLflow	- Implement a variety of tasks within a job - Using notebooks, JARS, Delta Live Tables pipelines, or Python, Scala, Spark submit, SQL and Java	- ETL processes - Compatible with batch and streaming inputs - Enforced data quality and consistency. - Tracking & logging of data transformation
Orchestration of Dependent Jobs	Machine Learning Tasks	Arbitrary Code, External API Call, Custom Tasks	Data Ingestion and Transformation

Note: DLT pipeline can be a task in a workflow



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- **Two main services = Workflow Jobs & Delta Live Tables (DLT)**
 - We cover DLT in ML Deployment (see self paced learning for free if not take it)
- **Job = Orchestration of Dependent Jobs, Machine Learning Tasks, Code/Custom Tasks**
 - **Jobs = more flexible**

Databricks has two task orchestration services;

The first one is the Workflow Jobs, which we will simply call it as Workflows in this lesson.

The second service is the Delta Live Tables (DLT). In this course, we are not going to cover Delta Live Tables.

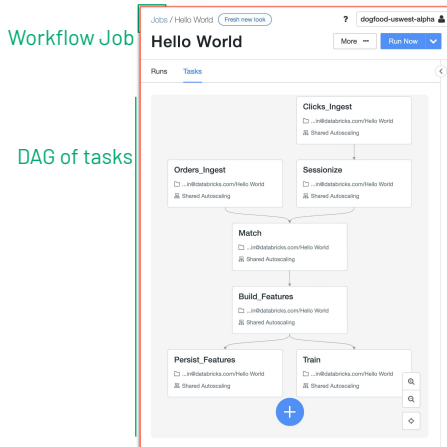
Workflows is the service for orchestrating for all types of tasks. In other words, it can orchestrate any combination of notebooks, SQL, Spark, ML models, and Delta Live Tables.

On the other hand, Delta Live Tables is designed for data teams to easily create batch and streaming data pipelines using Python or SQL. DLT has built in quality controls and data quality monitoring which ensures accurate and useful Business Intelligence, Data Science, and Machine learning built on top of quality data.

It is important to note that while these two services serve different purposes, they can be easily integrated. For example, a DLT pipeline can be executed as a task in a Workflow.

Workflows Jobs

Key Features



- **Easy creation, scheduling, and orchestration of your code with a DAG** (Directed Acyclic Graphs)
- Key features
 - **Simplicity:** Easy creation and monitoring in the UI
 - **Many Tasks** types suited to your workload
 - **Fully integrated** in Databricks platform, making inspecting results, debugging faster
 - **Reliability** of proven Databricks scheduler
 - **Observability** to easily monitor status

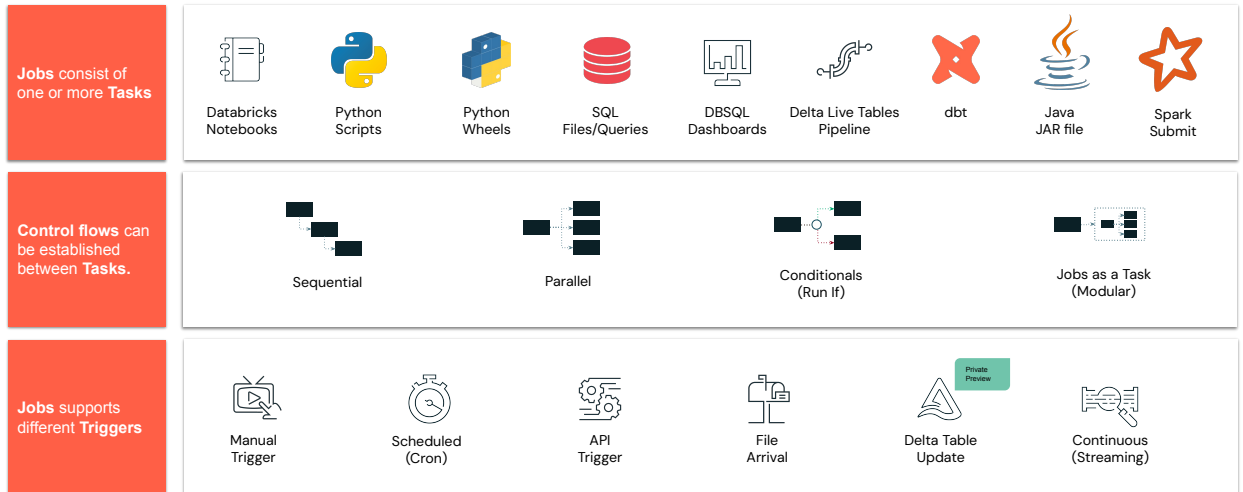


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- **Simplifies the orchestration** of workflows with a user friendly interface and **streamlined configuration, scheduling, and management using Directed Acyclic Graphs (DAGs)**.
 - **Monitoring and Observability**
 - i. Empowers users with tools to actively **track the status and performance** of each job.
 - ii. Provides **insights into job execution**, aiding in proactive management and adjustment of workflows.
 - Built-in mechanisms for **task retry and failover**
- **Decompose large-scale processes** into smaller, manageable units that can **operate in sequence or concurrently**.
 - Allows users to establish clear **task dependencies**

Building Blocks of Databricks Workflows Job

A unit of orchestration in Databricks Workflows is called a Job.

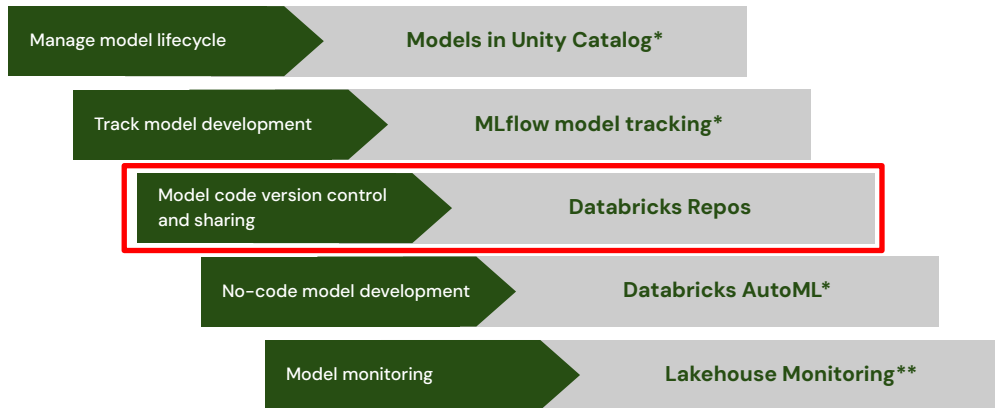


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- **Diverse Task Compatibility:** Supports a broad spectrum of tasks, tailored to meet various operational demands across data environments.
- **Flexible flows**
- **Wide variety of triggers**
 - *Delta Table Update = Trigger in Private Preview*
 - *Private Preview Express Interest Form:*
<https://docs.google.com/forms/d/e/1FAIpQLScbcQlcvliZCumlQgjHD0IC3y9bFM8tPAfBMFUyxhJpBcrdFg/viewform>

ModelOps Tasks and Tools in Databricks

The table lists common ModelOps tasks and tools provided by Databricks:



* Discussed in other associated machine learning courses.

** Covered later in the training

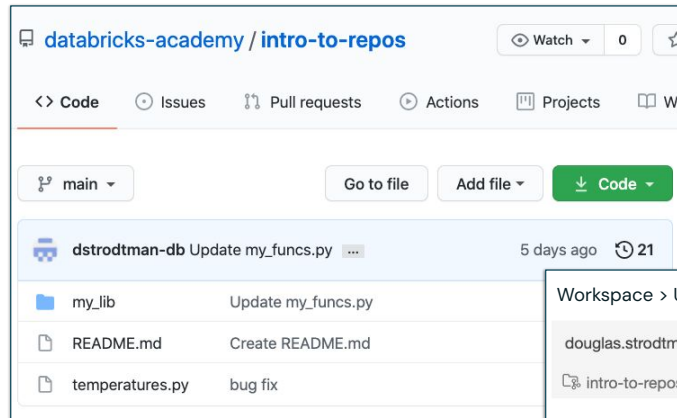


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- **ModelOps**, in Databricks streamline the lifecycle management of machine learning models, from development to deployment and monitoring
- **Databricks' offerings** support models that are not only developed with precision but are also maintained effectively throughout their operational life
- **The table lists common ModelOps tasks and tools in Databricks:**

Introduction to Git Folders

Git folders provide a **visual Git client & API** within Databricks, allowing users to **manage** code repositories, **collaborate**, and **integrate** with Git services.



Key Features:

- Seamless Git integration.
- Collaborative coding environment.
- Simplified version control.

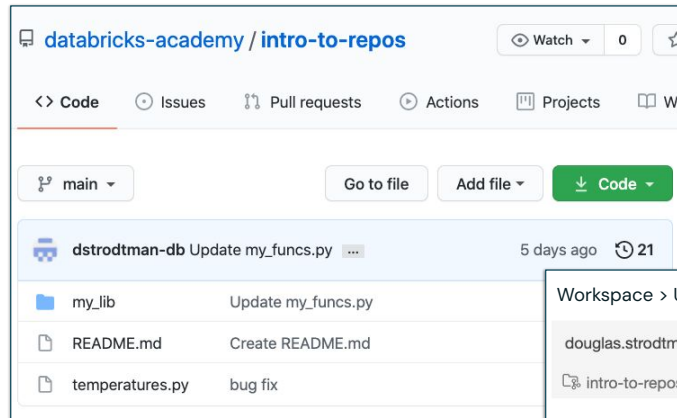


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- **Databricks Repos** serve as a **visual Git client and API** within Databricks.
- **Key Features:**
 - **Seamless Git Integration:**
 - Easily connect and manage repositories from popular Git providers, integrated directly within Databricks for a unified experience.
 - **Collaborative Coding Environment:**
 - Develop code in notebooks or other files collaboratively.
 - Use Git for version control and CI/CD.
 - **Simplified Version Control:**
 - Maintain a history of changes
 - Utilize visual tools for comparing changes and managing branches

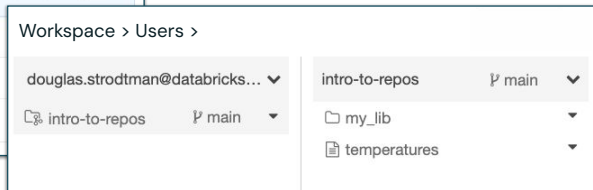
Introduction to Git Folders

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Notebooks and directories within the remote repository will be synced back to the workspace.

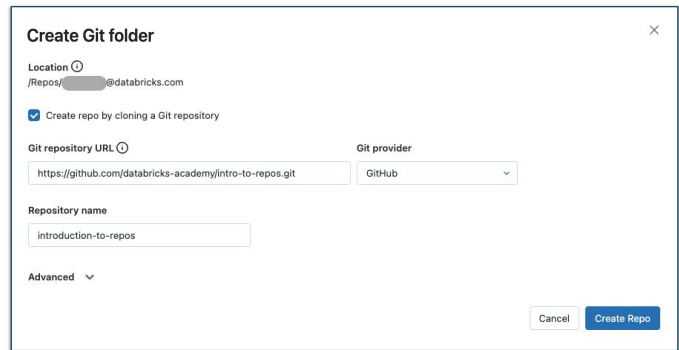
This functionality makes it easier for more users to leverage the power of Git in their development processes without the workarounds necessary in previous iterations of the Databricks platform.

Repos adds support to Databricks for branching and commit comments, making it easy to push changes back to your remote repository and kick off the next step in your development and deployment lifecycle.

Git Folder Setup and Commands

Once setup easily manage and perform Git operations within Databricks

- From the Databricks UI common Git operations:
 - Clone
 - Checkout
 - Commit
 - Push
 - Pull
 - Branch management
- Uses Personal Access Token or equivalent to authenticate.



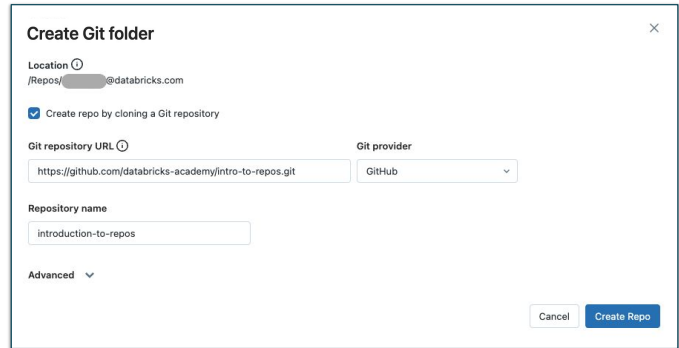
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- Supports common Git operations:
 - **Cloning** repositories
 - **Committing and pushing** changes
 - **Pulling** updates
 - **Branch management**
- Databricks Git folders uses a personal access token (PAT) or an equivalent credential to authenticate with your Git provider to perform operations
 - **Cloud Git Providers:**
 - GitHub, GitHub AE, and GitHub Enterprise Cloud
 - Atlassian BitBucket Cloud
 - GitLab and GitLab EE
 - Microsoft Azure DevOps (Azure Repos)
 - AWS CodeCommit

Git Folder Setup and Commands

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- **On-Premises Git Providers:**
 - GitHub Enterprise Server
 - Atlassian BitBucket Server and Data Center
 - GitLab Self-Managed
 - Microsoft Azure DevOps Server

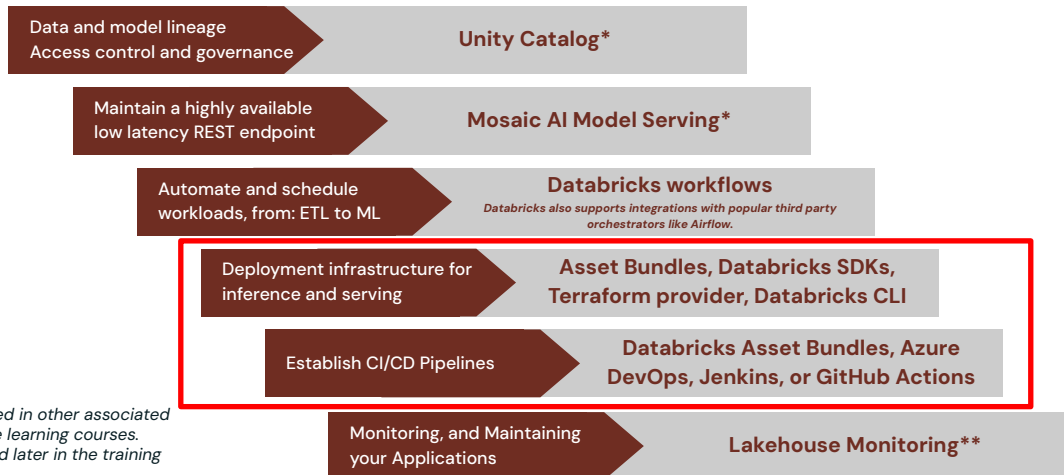
Visual comparison of diffs when committing

Repos is an extension of the Databricks Data Science & Engineering workspace. It maintains the pattern of private directories for individual users or shared directories with customize access controls.

What it adds is a simple UI for connecting to a remote Git repository, enabling project-level control over Databricks notebooks.

DevOps: Production and automation

The table lists common DevOps tasks and tools provided by Databricks:



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machine learning courses.

** Covered later in the training



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Instructor Notes:

- **DevOps within Databricks** focuses on enhancing the collaboration, automation, and monitoring of software development processes to streamline the lifecycle of data analytics and machine learning applications.
 - leverages a blend of practices and tools that help teams **accelerate development cycles** while ensuring high-quality and stable deployments

Overview Developer Tools

Databricks Asset Bundles (Recommended)

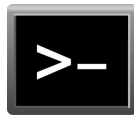
- Infrastructure as Code for Databricks resources and workflows



Databricks
Asset
Bundles

Databricks CLI

- Easy-to-use interface for automation from terminal, command prompt, or bash scripts.



Databricks SDKs

SDKs for:

- Python
- Java
- Go
- R



Databricks REST API

- Universal way to interact with Databricks workspaces: start/stop clusters, trigger jobs, manage models, update permissions, etc.



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- **laC = Asset Bundles** for defining and managing Databricks resources and workflows as code, ensuring consistent and repeatable deployments.
- Tailored specifically to enhance the management of Databricks environments, simplifying complex configurations and integrations.
- **Databricks Command Line Interface (CLI)**
 - Offers a straightforward command-line interface that simplifies automation of Databricks tasks from terminal or script environments like Bash.
 - **Ideal for developers and system administrators** looking to script interactions with Databricks
- **Databricks Software Development Kits (SDKs)**
 - **Allows developers to programmatically interact with Databricks services**, supporting everything from data manipulation to ML model deployment.
 - Provides SDKs for Python, Java, Go, and R, facilitating development across different technology stacks.
- **Terraform Provider for Databricks**
 - Utilizes Terraform to **manage not only Databricks workspaces** but also associated cloud infrastructure

What is a DAB?

- **Databricks Asset Bundles or DABs** are a collection of Databricks **artifacts** ¹ (e.g. jobs, ML models, DLT pipelines, and clusters) and **assets** ² (e.g. Python files, notebooks, SQL queries, and dashboards).
- These DABs (aka **bundles**) are **configured through YAML files** and can be **co-versioned** in the same repository as the assets and artifacts referenced in the bundle.
- Using the **Databricks CLI** these bundles can be materialized across multiple workspaces **like dev / staging and production** enabling customers to integrate these into their **automation** and **CI/CD** processes

1 Assets are instantiations of sources that persist state.

2 Artifacts are file-like resources that exist on a workspace path that carry little or no state



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Databricks Asset Bundles (DABs) apply Infrastructure as Code (IaC) principles to manage and deploy Databricks resources such as jobs, pipelines, notebooks, Delta Live Tables, Model Serving endpoints, and MLflow experiments. Each bundle contains cloud and workspace configurations, resource definitions, source files like notebooks and Python scripts, as well as unit and integration tests to ensure reliability. By aligning with software engineering best practices, DABs facilitate collaboration on data, analytics, and ML projects through standardized environments, enable faster iterations with streamlined updates and rollbacks, and maintain a versioned history of code and infrastructure for compliance. They are particularly suited for continuous integration and deployment workflows and allow organizations to create custom bundle templates that enforce consistency and standardization across projects.



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